

Analysis Leaving Out Donor 8

2026-06-07

Fig 6A, 6B, and 6C

Data is log10 transformed for 6B and 6C and not 6A.

Question: Determine if there is a statistically significant difference between asymptomatic and severe donors at each dose (6A, 6B) or at each time (6C) for each gene (CXCL10, CCL5, IL15, IL12, IFI16) at dose 100.

Data represents three independent biological replicates for all 10 donors.

We first analyzed the data with linear mixed models. There are some warnings about singular values in 6B and 6C. Hence we re-analyzed the data with unpaired t tests.

Table 1: Linear mixed models.

6A	0	0.1	1	10	100
CXCL10	0.17 (CI=-1.27, 1.60, p=0.820)	3.61 (CI=0.83,6.38, p=0.011)*	2.80 (CI=0.90,4.70, p=0.004)**	1.87 (CI=0.31,3.43, p=0.019)*	1.12 (CI=-0.81,3.04, p=0.254)
CCL5	-0.38 (CI=-1.48, 0.73, p=0.502)	-0.18 (CI=-1.31, 0.96, p=0.761)	0.31 (CI=-0.91, 1.52, p=0.622)	-0.20 (CI=-1.54, 1.14, p=0.771)	-0.39 (CI=-1.44, 0.65, p=0.461)
IL15	0.43 (CI=-1.21, 2.07, p=0.608)	0.51 (CI=-0.73,1.75, p=0.422)	0.71 (CI=-0.28,1.69, p=0.160)	0.34 (CI=-0.48,1.17, p=0.416)	0.18 (CI=-0.42,0.78, p=0.565)
IL12A	-0.40 (CI=-1.42, 0.62, p=0.444)	0.30 (CI=-1.09,1.68, p=0.674)	0.34 (CI=-0.77,1.46, p=0.548)	-0.22 (CI=-1.35,0.90, p=0.699)	-0.20 (CI=-1.21,0.82, p=0.705)
IFI16	-0.14 (CI=-1.07,0.80, p=0.775)	0.10 (CI=-0.58,0.78, p=0.772)	0.38 (CI=-0.25,1.00, p=0.234)	0.25 (CI=-0.45,0.95, p=0.482)	-0.20 (CI=-0.86,0.47, p=0.564)

Table 2: Unpaired t tests.

6A	0	0.1	1	10	100
CXCL10	0.817	0.033	0.021	0.123	0.357
CCL5	0.543	0.779	0.655	0.791	0.494
IL15	0.632	0.449	0.219	0.451	0.611
IL12A	0.435	0.698	0.581	0.699	0.702
IFI16	0.804	0.790	0.321	0.572	0.581

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Table 3: Linear mixed models.

6B	0	0.1	1	10	100
CXCL10	0.00 (CI=-0.13,0.13, p=0.999)W	-1.04 (CI=-1.84,-0.24, p=0.011)*	-0.79 (CI=-1.35,-0.23, p=0.006)**	-0.51 (CI=-1.00,-0.03, p=0.038)*	-0.29 (CI=-0.86,0.29, p=0.326)
CCL5	0.00 (CI=-0.16,0.16, p=0.999)W	-0.06 (CI=-0.22,0.10, p=0.465)W	-0.21 (CI=-0.40,-0.02, p=0.032)*	-0.05 (CI=-0.33,0.23, p=0.710)	0.00 (CI=-0.19,0.20, p=0.959)
IL15	0.01 (CI=-0.09,0.12, p=0.824)W	-0.01 (CI=-0.47,0.44, p=0.961)	-0.07 (CI=-0.54,0.40, p=0.771)	0.04 (CI=-0.24,0.32, p=0.791)	0.09 (CI=-0.24,0.42, p=0.600)
IL12A	0.01 (CI=-0.11,0.14, p=0.851)W	-0.20 (CI=-0.56,0.17, p=0.290)	-0.21 (CI=-0.49,0.07, p=0.135)	-0.04 (CI=-0.19,0.10, p=0.577)	-0.05 (CI=-0.19,0.09, p=0.481)W
IFI16	0.00 (CI=-0.12,0.11, p=0.992)W	-0.07 (CI=-0.27,0.12, p=0.468)	-0.16 (CI=-0.39,0.08, p=0.195)	-0.12 (CI=-0.28,0.05, p=0.164)	0.02 (CI=-0.14,0.17, p=0.821)

Table 4: Unpaired t tests.

6B	0	0.1	1	10	100
CXCL10	0.812	0.042	0.024	0.078	0.360
CCL5	0.905	0.394	0.060	0.710	0.957
IL15	0.373	0.962	0.782	0.805	0.622
IL12A	0.367	0.346	0.201	0.636	0.497
IFI16	0.102	0.526	0.236	0.186	0.845

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Table 5: Linear mixed models.

6C	Untreated	4HPT	24HPT
CXCL10	0.00 (CI=-0.13,0.13, p=0.999)W	-0.29 (CI=-0.86,0.29, p=0.326)	0.19 (CI=-0.51,0.90, p=0.588)
CCL5	0.00 (CI=-0.16,0.16, p=0.999)W	0.00 (CI=-0.19,0.20, p=0.959)	-0.06 (CI=-0.64,0.53, p=0.847)
IL15	0.01 (CI=-0.09,0.12, p=0.824)W	0.09 (CI=-0.24,0.42, p=0.600)	0.05 (CI=-0.47,0.58, p=0.847)
IL12A	0.01 (CI=-0.11,0.14, p=0.851)W	-0.05 (CI=-0.19,0.09, p=0.481)W	-0.15 (CI=-0.54,0.25, p=0.460)
IFI16	0.00 (CI=-0.12,0.11, p=0.992)W	0.02 (CI=-0.14,0.17, p=0.821)	-0.01 (CI=-0.39,0.37, p=0.957)

Table 6: Unpaired t tests.

6C	Untreated	4HPT	24HPT
CXCL10	0.812	0.360	0.574
CCL5	0.905	0.957	0.843
IL15	0.373	0.622	0.842
IL12A	0.367	0.497	0.451
IFI16	0.102	0.845	0.956

Fig 7A and 7B

Question: Determine if there is a statistically significant difference between the expression level of each gene (CXCL10, CCL5, IL15, IL12, IFI16) at each dose compared to the previous dose. For clarity, this is comparing 0.0 to 0.1, 0.1 to 1, 1 to 10, and 10 to 100. Data represents three independent biological replicates from 10 different donors

Data is log10 transformed.

Table 7: Paired t tests.

7A	0.1 vs. 0	1 vs. 0.1	10 vs. 1	100 vs. 10
CXCL9	<0.001	<0.001	<0.001	0.008
CXCL10	<0.001	<0.001	0.004	0.619
CXCL11	<0.001	<0.001	0.002	0.020

Question: Determine if there is a statistically significant difference between asymptomatic and severe donors at each dose (0, 0.1, 1, 10, 100). Data represents three independent biological replicates for all 10 donors.

Data is log10 transformed.

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Table 8: Linear mixed models.

7B	0	0.1	1	10	100
CXCL9	0.00 (CI=-0.30,0.30, p=1.000)W	-1.38 (CI=-2.58,-0.17, p=0.025)*	-0.97 (CI=-1.88,-0.07, p=0.035)*	-0.98 (CI=-1.45,-0.51, p=<0.001)**	-0.75 (CI=-1.19,-0.31, p=0.001)**
CXCL10	0.00 (CI=-0.13,0.13, p=0.999)W	-1.04 (CI=-1.84,-0.24, p=0.011)*	-0.79 (CI=-1.35,-0.23, p=0.006)**	-0.51 (CI=-1.00,-0.03, p=0.038)*	-0.29 (CI=-0.86,0.29, p=0.326)
CXCL11	0.00 (CI=-0.25,0.25, p=1.000)W	-1.02 (CI=-2.27,0.22, p=0.107)	-0.95 (CI=-1.86,-0.04, p=0.041)*	-0.58 (CI=-1.23,0.08, p=0.084)	-0.35 (CI=-1.09,0.39, p=0.355)

Table 9: Unpaired t tests.

7B	0	0.1	1	10	100
CXCL9	0.555	0.077	0.077	0.007	0.025
CXCL10	0.812	0.042	0.024	0.078	0.360
CXCL11	0.907	0.195	0.122	0.160	0.401

Updated Luminex data

Data is log10 transformed.

Question: Is there a difference between asymptomatic and severe donors at each dose in each cytokine? IE 0U vs 0U, 1U vs 1U, 100U vs 100U.

Table 10: Linear mixed models. P values set to N/A for CXCL9 and CXCL11 at 0U because all values are under LOD and they only differ in concentration factors.

	0U	1U	100U
CXCL9	NA	-0.25 (CI=-0.72,0.22, p=0.294)	-0.40 (CI=-0.90,0.11, p=0.122)
CXCL10	-0.40 (CI=-1.12, 0.32, p=0.275)	-0.48 (CI=-0.77,-0.19, p=0.001)**	-0.44 (CI=-0.79,-0.09, p=0.013)*
CXCL11	NA	-0.40 (CI=-1.15,0.35, p=0.293)	-0.82 (CI=-1.61,-0.03, p=0.041)*

Table 11: Unpaired t test pvals. P values set to N/A for CXCL9 and CXCL11 at 0U because all values are under LOD and they only differ in concentration factors.

	0U	1U	100U
CXCL9	NA	0.319	0.195
CXCL10	0.351	0.025	0.062
CXCL11	NA	0.326	0.104

Question: Is there a difference between 0U and 1U for each chemokine? Between 0U and 100U? Between 1U and 100U?

Table 12: Paired t test p values for asym.

	0U vs. 1U	0U vs. 100U	1U vs. 100U
CXCL9	0.016	<0.001	<0.001
CXCL10	<0.001	<0.001	<0.001
CXCL11	0.010	<0.001	<0.001

Table 13: Paired t test p values for severe.

	0U vs. 1U	0U vs. 100U	1U vs. 100U
CXCL9	0.051	0.003	<0.001
CXCL10	0.004	<0.001	<0.001
CXCL11	0.054	0.003	<0.001